



BICOL UNIVERSITY
COLLEGE OF SCIENCE
Legazpi City, Albay
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Individual Journal

Name: Jose Manuel G. Labalan

Group Name: Project Bloom

Role in the Group: Project Manager, Backend Developer, Tester

Topic / Activity: Laboratory 4: REST API

Date Submitted: April 16, 2026

Overview of My Role

As the group's leader, I mostly focused on giving tasks, managing team roles, and overseeing the project's big picture. I conceptualized the initial database structure and contributed to the backend API development. By keeping a high-level view of the system, I ensured that the overall architecture made sense and that everyone's individual parts would eventually come together seamlessly.

Specific Tasks Performed

- I formulated the initial idea and relational structure for the database, determining how users, students, and subjects should interact within the GradeTrack system.
- I managed the assignment of roles, delegated tasks, and maintained a high-level overview of each member's progress to ensure our deliverables met the activity requirements.
- I collaborated on the backend PHP development, helping structure how the API receives, parses, and responds to client requests.
- I actively tested the system, specifically verifying that the data flow from the database through the API was functioning as intended.

Code Contributions

- I drafted the initial concepts and relational structure for the database, specifically designing how the users, students, and subjects tables would interact.
- In the backend, I contributed to the PHP development by structuring the HTTP method routing logic within `api.php`.
- I wrote the foundational routing logic that determines how the API handles and directs incoming GET, POST, PUT, and DELETE requests, ensuring the server could accurately parse client commands before passing them off to the core endpoints.

Challenges Encountered

The main challenge I encountered was maintaining synchronization between the frontend and backend teams. Because multiple members were touching different parts of the system simultaneously (like the API, the Python CLI, and the web client), there was a high risk of code conflicts. Furthermore, it was difficult to ensure that the data structure I originally conceptualized for the backend would perfectly align with what the frontend developers needed to display.

Problem-Solving and Contributions

To resolve the synchronization issues and minimize code conflicts, I strictly managed the team's roles and acted as the primary bridge between the frontend and backend developers and made sure everyone is having the same idea. Whenever there was a disconnect between what the web client needed and what the API was providing, I stepped in to clarify the requirements. Furthermore, to guarantee that the frontend could properly consume our data, I continuously tested the API responses against the database and acted like a “baby” and tried multiple scenarios on the program.